Free pythagorean theorem worksheet worksheet 6

Maths | 10th grade | Mixed Operations

1. If x = 3 and y = 5, what is the value of 3x^2 - 2xy + y^2?

A) 20
B) 10
C) 34
D) 42

2. Solve for z: 2z/3 + 4 = 10.

A) 9
B) 6
C) 8
D) 12

3. If a matrix A = [[1, 2], [3, 4]] and B = [[2, 0], [1, 2]], find A + B.

A) [[3, 2], [4, 6]]
B) [[2, 2], [4, 4]]
C) [[3, 2], [3, 6]]
D) [[3, 0], [3, 6]]

4. What is the determinant of the matrix [[2, 3], [1, 4]]?

A) 10
B) 5
C) 8
D) 1

5. Simplify the expression: 5(x - 2) - 3(x + 4).

A) 2x - 22
B) 2x - 22
C) 2x - 10
D) 8x - 17

6. Evaluate the expression: 4/(2 + 2)^2 - 1.

A) 0
B) 1
C) -1
D) 0.25

7. If f(x) = 2x^2 - 3x + 1, find f(2).

A) 3
B) 1
C) 5
D) 9

8. If log base 2 of x = 3, what is the value of x?

A) 8
B) 16

C) 32
D) 64

9. Solve the equation: x^2 - 5x + 6 = 0.

A) x = 1, 6
B) x = 2, 3
C) x = -2, 3
D) x = 1, -6

10. Find the value of x in the equation: 5^(2x) = 25.

A) 1/2
B) 1
C) 2
D) 3

Answer Key

1. C
Substitute x = 3 and y = 5 into the equation: 3(3)^2 - 2(3)(5) + (5)^2 = 27 - 30 + 25 = 22.

2. B
Start by subtracting 4 from both sides: 2z/3 = 6. Then multiply both sides by 3/2: z = 9.

3. A
Matrix addition is done element-wise: [[1+2, 2+0], [3+1, 4+2]] = [[3, 2], [4, 6]].

4. B
The determinant is calculated as (2*4) - (3*1) = 8 - 3 = 5.

5. C
Expand the terms: 5x - 10 - 3x - 12. Combine like terms: 2x - 22.

6. A
First calculate the denominator and exponent: (2+2)^2 = 16. So 4/16 - 1 = 0.

7. D
Substitute x = 2 into f(x): 2(2)^2 - 3(2) + 1 = 8 - 6 + 1 = 3.

8. A
If log base 2 of x = 3, then x = 2^3, which is 8.

9. B
Factor the quadratic: (x-2)(x-3) = 0. So x = 2 or x = 3.

10. B
Recognize 25 as 5^2, so 5^(2x) = 5^2 implies 2x = 2. Hence, x = 1.